attacker that gains privileged access to observe all network communications within the cluster by default. They can use credentials discovered from this snooping to access other systems and increase the scope of the attack.

> DEFINITION ***Network segmentation*** refers to using switches, routers, and firewalls to divide a network into separate *segments* (also known as *collision domains*). An attacker can then only observe network traffic within the same network segment and not traffic in other segments.

Although there are approaches that provide some of the benefits of segmentation within a cluster, a better approach is to actively protect all communications using TLS. Apart from preventing an attacker from snooping on network traffic, TLS also protects against a range of attacks at the network level, such as the DNS rebind attacks mentioned in section 10.2.8. The certificate-based authentication built into TLS protects against spoofing attacks such as *DNS cache poisoning* or *ARP spoofing*, which rely on the lack of authentication in low-level protocols. These attacks are prevented by firewalls, but if an attacker is inside your network (behind the firewall) then they can often be carried out effectively. Enabling TLS inside your cluster significantly reduces the ability of an attacker to expand an attack after gaining an initial foothold.

> DEFINITION In a ***DNS cache poisoning attack***, the attacker sends a fake DNS message to a DNS server changing the IP address that a hostname resolves to. An ***ARP spoofing attack*** works at a lower level by